

■ Agentic UAV Mission Planning and Safety Compliance Assistant User Manual

Agentic UAV Mission Planning and Compliance Auditor

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1. System Introduction

Agentic UAV Mission Planning and Safety Compliance Assistant is a software assistant designed to planning UAV autonomous operations, visualize dynamic flight trajectories, and verify safety rule compliance. The system uses a multi-agent AI architecture to translate natural language requirements, generate route waypoints, checks no-fly zones, and suggest optimal parameter corrections before takeoff.

1.1 Core Architecture & Agents

The system is divided into five modular agents cooperating in sequence:

- **Mission Understanding Agent:** Parses natural language requests using Google Gemini Generative AI models.
- **Waypoint Planner Agent:** Calculates geometric paths (Square, Grid scanning, Circle orbit, Perimeter).
- **Safety Compliance Agent:** Enforces 7 compliance safety rules using physics and Shapely metrics.
- **Correction Agent:** Adjusts flight coordinates and clips bounds in response to failed checks.
- **Report Generation Agent:** Compiles detailed PDFs, CSV logs, and QGC plan files.

2. System Setup & Running Instructions

Agentic UAV Mission Planning and Safety Compliance Assistant requires Python 3.8+ and standard dependencies (Streamlit, Folium, Shapely, ReportLab, etc.).

2.1 Installation Steps

Follow these terminal steps to launch the application locally:

```
git clone https://github.com/AbdulAzeemHashmi/agentic-uav-mission-planner.git
cd agentic-uav-mission-planner
python -m venv .venv
.venv\Scripts\Activate # On Windows
pip install -r requirements.txt
streamlit run app.py
```

3. Airspace Compliance Safety Controls

The system enforces 7 crucial safety standards to prevent UAV crashes and airspace regulations violations:

Rule ID	Standard Limit	Compliance Implementation Detail
R1	Altitude Ceiling $\leq 80\text{m}$	Enforces legal flying boundaries
R2 / R3	Takeoff & Landing/RTL	Ensures terminal control nodes
R4	Geofence Clearance	Checks waypoint and line-segment intersections
R5	Leg Length $\leq 500\text{m}$	Limits max distance between waypoints
R6	Flight Window $\leq 30\text{ min}$	Caps total operational duration
R7	Battery Consumption $< 80\%$	Uses dynamic physics energy calculation

4. Geofence Corrections & Telemetry Exporting

Agentic UAV Mission Planning and Safety Compliance Assistant implements mathematical projections to resolve compliance issues and formats outputs for real controllers:

4.1 Shapely Boundary-Normal Corrections

If a waypoint or path segment violates restricted airspace, the Correction Agent does not perform a simple offset. It projects the coordinate to the nearest polygon boundary point using Shapely algorithms and shifts it along the outward normal vector by exactly 15.0 meters. For circular no-fly zones, coordinates are shifted along the center-outward radial vector.

4.2 Physics-Based Battery Estimation

Battery consumption (R7) computes climb energy (215W at 4m/s), cruise energy (120W at 10m/s), descent energy (115W at 2m/s), and residual hovering states (130W) based on UAV mass (1.5 kg) and a 90 Wh capacity model.

4.3 QGroundControl (.plan) File Export

You can export flight plans as standard JSON .plan files. These contain mapped MAVLink commands (MAV_CMD_NAV_TAKEOFF, MAV_CMD_NAV_WAYPOINT, MAV_CMD_NAV_RETURN_TO_LAUNCH) which are directly importable into QGroundControl or standard drone autopilot systems.